



ASSIGNMENT 1

[100 MARK]

INSTRUCTION

- This assignment is a combination of **Chapter 1 to Chapter 5**.
 - Please **answer the question in the A4 paper**.
 - Your answer must be **handwritten**, and all the solution must be shown.
 - Make sure your handwriting is easily to be understood.
 - Please print out your Microsoft Excel solution and attach it to your assignment submission. Make sure all data and calculations are visible
 - Make sure to **print out and attach a cover paper to the assignment submitted**.
 - This assignment must be submitted **individually**.
 - **Attempt of cheating** will directly give you **0 Mark** for your assignment.
 - The use of AI will lead to deduction of 10 marks for every question.
 - Late submission of assignment might lead to **deduction of 10 mark per day**.
 - **SUBMISSION DATE: 14 OGOS 2026 (Before 4pm)**
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QUESTION 1 [32 Marks]

The U.S. National Center for Health Statistics published data on weight and heights by an age and sex in the document Vital and Health Statistics. The weight is shown in the Table below, given the nearest tenth of a pound, were obtained from sample of 18-24 years old males.

129.2	185.3	218.1	187.5	145.6
155.2	170.0	151.3	187.5	172.5
167.3	161.0	178.7	165.0	178.2
191.1	150.7	187.0	173.7	136.7
161.7	170.1	165.8	214.6	158.5
278.8	175.6	188.7	132.1	173.6
146.4	209.1	175.4	182.0	188.7
149.9	158.6	182.5	142.8	209.1

- a. Why do you think the researchers selected a sample instead of measuring every 18–24-year-old male in the United States?

[2 Marks]

- b. The data were collected only from males aged 18–24 years. Can the findings be generalized to females or older adults? Why or why not?

[2 Marks]

- c. Develop a frequency distribution with 7 classes. Make a tally mark for each data entry to find the total frequency, f for each class.

[4 Marks]

- d. Calculate the midpoint, relative frequency and cumulative frequency for each class.

[5 Marks]

- e. Draw a frequency histogram and ogive of the weight in pounds of 40 males.

[10 Marks]

- f. Calculate the sample mean, sample standard deviation.

[6 Mark]

- g. Coefficient variation of the weight in pounds of 40 males. Interpret the value.

[3 Mark]

QUESTION 2 [18 Marks]

The following are the published weights (in pounds) of all the members of the San Francisco 49ers football team from a previous season.

177	205	210	210	232	205	185
185	178	210	206	212	184	174
185	242	188	212	215	247	241
223	220	260	245	259	278	270
280	295	275	285	290	272	273
280	285	286	200	215	185	230
286	200	215	185	230	250	241
250	241	190	260	250	302	265
290	276	228	265			

By using Microsoft Excel software, key-in the data and solve the question given.

(Note: Make sure to print out your solution in Microsoft Excel and attach it to your submission).

- a. Arrange the data in ascending order.

[1 Mark]

- b. Find the following:

- i. Median
- ii. First Quartile (Q_1)
- iii. First Quartile (Q_3)
- iv. Determine the interquartile range (IQR) of the weights.

[4 Marks]

- c. Construct a box plot for the data.

[3 Marks]

- d. Based on your answer in part (b) describe the distribution of the data.

[2 Marks]

- e. Assume that the data represent the population of the San Francisco 49ers team.

Find:

- i. The population means.
- ii. The population standard deviation.

[5 Marks]

- f. Based on your answer in part (e), determine the data lying within $k = 3$ standard deviation. Interpret your answer.

[3 Marks]

QUESTION 3 [15 Marks]

A university is organizing a STEM Innovation Fair. Fourteen students have been shortlisted to participate. Their specialization is as follows:

Table 1: Distribution of students by specialization.

Specialization	Number of Students
Artificial Intelligence (AI)	5
Robotics (R)	4
Biotechnology (B)	3
Cybersecurity (C)	2

Each student belongs to exactly one specialization. The organizing committee must complete several tasks before the event. Answer the following questions.

- a. The committee wishes to appoint a Review Panel consisting of seven students. To ensure balanced representation, the panel must include:
- at least 2 AI students
 - at least 2 Robotics students, and
 - exactly 1 Cybersecurity student.

Determine the number of possible Review Panels.

[6 Marks]

- b. The selected Review Panel from part (a) will be seated in a single row for a group discussion. Suppose the chosen panel contains exactly
- 3 AI students,
 - 2 Robotics students,
 - 1 Biotechnology student, and
 - 1 Cybersecurity student.

Determine the number of seating arrangements if the Cybersecurity student must sit at one end of the row and the two Robotics students must not sit next to each other.

[4 Marks]

- c. The event programme records only the specialization of each participation rather than the students' names. Determine the number of distinct programme sequences that can be produced.

[5 Marks]

QUESTION 4 [16 Marks]

A publishing company estimates that **2%** of newly printed books contain a typographical error. An editor randomly checks books one by one.

- a. Find the mean, variance, and standard deviation.

[10 Marks]

- b. How many books are expected to be examined before finding one with a typographical error?

[2 Marks]

- c. On average, a printed books contains five typographical errors. Find the probability that a printed books contains exactly seven typographical errors. Interpret your answer.

[3 Marks]

- d. Based on your answer in part (c), state the type of probability distribution used.

[1 Mark]

QUESTION 5 [19 Marks]

Consider a population of eight students whose ages, in years, are those given in Table 5.8 (next page). Let X denote the age of a randomly selected student. Meanwhile, the probability distribution of the random variable X is shown in Table.

Table 2: Age and probability distribution of X for eight randomly selected students.

Age, X	f	$P(X = x)$
19	2	0.250
20	3	0.375
21	2	0.250
27	1	0.125

Express the mean age of the students in terms of the probability distribution of X .

[5 Mark]

Solve the question.

- a. Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT?

[7 Mark]

- b. How high must an individual score on the GMAT in order to score in highest 5%?

[7 Mark]